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$$f(x, y) = \begin{cases} \frac{x^2 y^2 (x + y)}{x^4 + y^4} & \text{als } (x, y) \neq (0, 0) \\ 0 & \text{als } (x, y) = (0, 0) \end{cases}$$

Continuïteit:

Stel $x = r \cos \theta$ en $y = r \sin \theta$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{r \rightarrow 0} \frac{(r^2 \cos^2 \theta)(r^2 \sin^2 \theta)(r \cos \theta + r \sin \theta)}{r^4 \cos^4 \theta + r^4 \sin^4 \theta} \\ &= \lim_{r \rightarrow 0} \frac{r^5 \cos^2 \theta \sin^2 \theta (\cos \theta + \sin \theta)}{r^4 (\cos^4 \theta + \sin^4 \theta)} \\ &= \lim_{r \rightarrow 0} r \left(\frac{\cos^2 \theta \sin^2 \theta (\cos \theta + \sin \theta)}{\cos^4 \theta + \sin^4 \theta} \right) \\ &\Rightarrow f \text{ is continu in } (0, 0) \end{aligned}$$

Partiële afgeleiden:

$$\begin{cases} \frac{\partial f}{\partial x}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(\lambda, 0) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \\ \frac{\partial f}{\partial y}(0, 0) = \lim_{\lambda \rightarrow 0} \frac{f(0, \lambda) - f(0, 0)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{0}{\lambda} = 0 \end{cases}$$

$\Rightarrow f$ is partieel afleidbaar in $(0, 0)$

Afleidbaarheid:

Stel $\bar{h} = (h_1, h_2)$

$$\begin{aligned} Df(\bar{0}, \bar{h}) &= \lim_{\lambda \rightarrow 0} \frac{f(\lambda h_1, \lambda h_2) - f(0, 0)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{(\lambda^2 h_1^2)(\lambda^2 h_2^2)(\lambda h_1 + \lambda h_2)}{\lambda^4 h_1^4 + \lambda^4 h_2^4} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^5 h_1^2 h_2^2 (h_1 + h_2)}{\lambda (\lambda^4 h_1^4 + \lambda^4 h_2^4)} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda^5 h_1^2 h_2^2 (h_1 + h_2)}{\lambda^5 (h_1^4 + h_2^4)} = \frac{h_1^2 h_2^2 (h_1 + h_2)}{h_1^4 + h_2^4} \\ &\Rightarrow f \text{ is afleidbaar in } (0, 0) \end{aligned}$$

Differentieerbaarheid:

$$\text{Voor differentieerbaarheid in } (0, 0) \text{ moet gelden: } Df(\bar{0}, \bar{h}) = \begin{pmatrix} \frac{\partial f}{\partial x}(0, 0) & \frac{\partial f}{\partial y}(0, 0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = 0$$

$$Df(\bar{0}, \bar{h}) = \frac{h_1^2 h_2^2 (h_1 + h_2)}{h_1^4 + h_2^4} \neq 0$$

$$\Rightarrow f \text{ is niet differentieerbaar in } (0, 0)$$

References